COM 5506 Social Network Analysis for Communication

L8 Network related Statistics Analysis

SEM B 2026

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Reminder

9	3/18	Lecture: Network data hypothesis testing Lab: UCINET introduction	
10	3/25	Workshop 2: Group project progress consultation	Bring your data set to class
11	4/1	**Reading week**	
12	4/8	Final test (Lecture 1 to Lecture 8) Workshop 3: Group project finding consultation	Bring your preliminary findings to class
13	4/15	**No Class**	
14	4/22	*Final Project Presentation Day*	Complete your project draft
	4/29		**Research paper due**

Project Report Format

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- Title
- Author list
- Abstract (about 100 words)
- Introduction
  - What is the project about (the purpose);
- Method (details)
  - How is the data collected (source, procedure, method, etc)
  - How is the data analyzed; (analysis tools, techniques, statistics)
  - How is the data visualized (visualization tools)

Project Report Format

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- Results/Findings
  - Data analysis
  - Table(s)
  - Visualizations/Figures
- Conclusions
- Appendix (data, code, etc.)
- Others
  - No need to attach references
  - No word count requirement; it's all about QUALITY, not about QUANTITY.
  - <u>Do it yourself</u>. DO NOT use labors from outside your group.

Final project

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- Explore a network and apply what you have learned (including knowledge from your Research Method class) to your project
  - You want to understand aspects of a real-world network
  - You want to understand underlying mechanisms shaping the network around us
  - You want to understand how features of this network are related to other non-network features?

1. Descriptive Analysis: network level

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- Network level features (to get to know the network)
  - Diameter -- meaning?
  - Average shortest path -- meaning?
  - Degree distribution -- meaning?
  - Density -- meaning?
  - Global clustering -- meaning?
  - Modularity -- meaning?
  - etc.
- All these statistics are at the network level. In other words, if you have one network, then you only have one number for each concept.

2. Descriptive Analysis: nodes level

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- Node level features (to get to know the individuals)
  - All kinds of centrality measures
    - Degree centrality
    - Betweenness centrality
    - Closeness centrality
    - Eigen value centrality
  - Clustering coefficient
  - All these are individual level data. In other words, each node will have one number for each indicator. Save them into an excel spread sheet or csv file, which could be further used for bivariate analysis.
  - Non-network node level features: don’t forget to NEED TO COLLECT data

3. Descriptive Analysis: edge level

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- Edge level features (to get to know the connections)
  - Edge weight being defined in the process of data collection
    - e.g., frequency based definition: if A and B are seen together for 5 times, then the edge weight for their connection is 5.
  - Edge weight are not available through data collection
    - e.g., your edge data does not have frequency based weight
    - You can calculate embeddedness for each edge and save data as weight
  - Edge weight distribution

4. Relationship Analysis: node level – network x non-network

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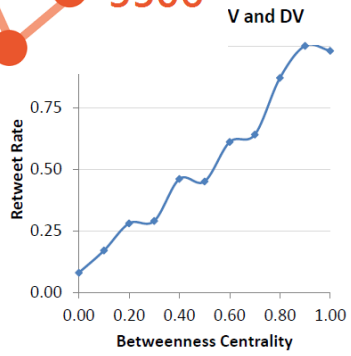

- Bivariate analysis (behavioral, semantic network features or other indicators not related to network but to individual nodes)
  - Node structure feature x node categorical attributes
    - e.g., centrality x gender. Man possess more important positions in the network?
    - Statistics: mean comparison, visualization: bar graph; no need to do significance test since you are not dealing with sample at all.
- Node structure feature x node continuous attributes
  - e.g., centrality x quiz grade, better grade students are more popular?
  - Statistics: Pearson's r
  - Visualization: two dimensional scatter plot; regression line plot
- All these could be done in excel, SPSS, or Python
- Grading principle: demonstrate to me what you learn from the course in terms of concepts, analysis, and software use techniques.

5. Relationship Analysis: node level – network x network

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- You can also correlate two network features of a node
  - e.g., degree x clustering
  - e.g., degree centrality x betweenness centrality
- All these could be done in excel, SPSS, or Python
- Grading principle: demonstrate to me what you learn from the course in terms of concepts, analysis, and software use techniques.

6. Relationship Analysis: network with network

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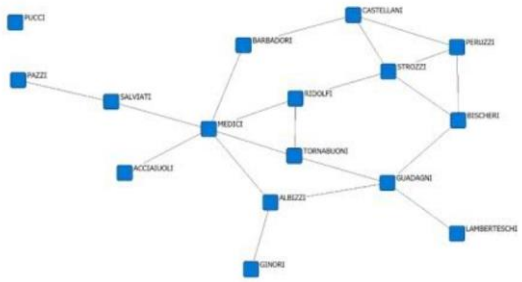
- How do two types of network correlated with each other?
- For example, do business and marriage relations influence each other? We have to use QAP correlation (Quadratic Assignment Procedure)
- Analysis prerequisite: networks grew out of the same nodes

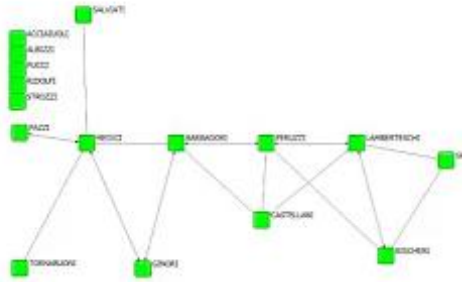
Independent variable

- Marriage network



Dependent variable

- Financial network



Data

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- Florentine Family Marriage and Business Ties Data as a “network" object
- Description
- This is a data set of marriage and business ties among Renaissance Florentine families. The data is originally from Padgett (1994) via UCINET and stored as a network object.
- Breiger & Pattison (1986), in their discussion of local role analysis, use a subset of data on the social relations among Renaissance Florentine families (person aggregates) collected by John Padgett from historical documents. The two relations are business ties (flobusiness - specifically, recorded financial ties such as loans, credits and joint partnerships) and marriage alliances (flomarriage).
- As Breiger & Pattison point out, the original data are symmetrically coded. This is acceptable perhaps for marital ties, but is unfortunate for the financial ties (which are almost certainly directed). To remedy this, the financial ties can be recoded as directed relations using some external measure of power - for instance, a measure of wealth. Both networks provide vertex information on (1) wealth each family's net wealth in 1427 (in thousands of lira); (2) priorates the number of priorates (seats on the civic council) held between 1282-1344; and (3) totalties the total number of business or marriage ties in the total dataset of 116 families (see Breiger & Pattison (1986), p 239).
- Substantively, the data include families who were locked in a struggle for political control of the city of Florence in around 1430. Two factions were dominant in this struggle: one revolved around the infamous Medicis (9), the other around the powerful Strozzis (15).

Lab 7

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- UCINET
  - http://www.analytictech.com/archive/ucinet.htm

- Data Management
  - Attribute data (cvs or excel)
  - Network data (network edge list)
  - Running analysis (SPSS or Excel or Python)
  - Visualization (Gephi or Python)

Lab 8 Putting everything together

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- Task 1 (network data collection): Using Python
  - Please construct a co-occurrence network based on the text files I uploaded onto Canvas: The Wonderful Wizard of Oz and Character name list for the novel. Co-occurrence is defined as two names (nodes) being mentioned in the same paragraph. Save the edge list into a csv file. Keep edge weight.
- Task 2 (non-network data collection): Using Excel
  - Skip – supposing I have collected some data for you already.
  - There is a csv file with non-network data features.
- Task 3 (visualization): Using Gephi
- Task 4 (network data descriptive analysis): Using either Python or Gephi
- Task 5 (network data correlational analysis): Using SPSS
  - Question: is friendliness score related to degree centrality?
  - Any gender difference in terms of degree centrality?
- Task 6 (network QAP correlational analysis): Using UCINET
  - Supposing we have another network based on the same character.
  - Character psychological distance to each other.
  - Test if the two networks are correlated.